Enteromorpha Prolifera Detection in High-Resolution Remote Sensing Imagery Based on Boundary-Assisted Dual-Path Convolutional Neural Networks

Zhipeng Dong, Yanxiong Liu, Yanli Wang, Yikai Feng, Yilan Chen, and Yanhong Wang

***Abstract*— Enteromorpha prolifera is a frequent marine ecological environment disaster. How to quickly and accurately monitor E. prolifera is of great significance to its management and protection of the marine ecological environment. The detection of E. prolifera from high spatial resolution remote sensing images (HSRIs) is an important technical means for monitoring E. prolifera disasters. With respect to the difficulty in accurate detection of E. prolifera area boundary in HSRIs, this article proposes an E. prolifera detection method for HSRIs based on boundary-assisted dual-path convolutional neural networks (BADP-CNNs). First, a large-scale HSRIs' E. prolifera detection dataset, FIO-EP, is created and published to facilitate the field of HSRIs' E. prolifera detection. Then, a BADP-CNN framework is designed to detect E. prolifera in HSRIs based on the shape distribution characteristics of E. prolifera. In the CNN framework, accurate detection of E. prolifera areas in HSRIs is achieved by fusing initial detection and boundary detection results of E. prolifera. The proposed method is compared with some state-of-the-art E. prolifera detection algorithms using the FIO-EP dataset. The experimental findings demonstrate that the proposed method can obtain 88.28% F1-score and 79.02% intersection-over-union (IOU) and is superior to other state-of-the-art E. prolifera detection algorithms.**

***Index Terms*— Convolutional neural networks (CNNs), deep learning, Enteromorpha prolifera detection, high spatial resolution remote sensing image (HSRI), multifeature fusion.**

I. Introduction

ENTEROMORPHA prolifera is a kind of *Ulva* family green algae living in saline and brackish water [1]. When a large amount of Enteromorpha prolifera accumulates, it will form catastrophic consequences, blocking sunlight, changing underwater light and biochemical conditions, affecting the growth and activities of marine organisms, and even causing marine disasters such as green tide outbreaks [2], [3], [4]. In addition to destroying the balance of marine ecosystem, E. prolifera disasters also cause huge losses to coastal fisheries, tourism, and harbor projects, as shown in Fig. 1 [5], [6], [7]. Therefore, how to quickly and accurately detect E. prolifera is of great significance to E. prolifera disaster management. Due to the large spatial distribution and drifting characteristics of E. prolifera, the traditional shipborne disaster detection system is difficult to effectively monitor E. prolifera [8]. However, the high-resolution remote sensing satellite has the characteristics of wide observation range, short revisit period, and high image resolution, which is very suitable for E. prolifera disaster monitoring [9]. With respect to the detection of E. prolifera in high spatial resolution remote sensing images (HSRIs), some researches have been carried out [10], [11], [12].

Based on the feature generation mode of remote sensing image E. prolifera detection, E. prolifera detection methods for remote sensing images are classified into two categories: traditional E. prolifera detection method and E. prolifera detection method based on convolutional neural networks (CNNs). Among them, the traditional E. prolifera detection method requires manually designed features to detect E. prolifera in remote sensing images. For example, Zhang et al. [13] used top-of-atmosphere reflectance (TAR) without any atmospheric correction scheme based on Tasseled-cap-like transformation to detect E. prolifera areas in huanjing-charge-coupled device (HJ-CCD) images. Cao et al. [14] applied the normalized difference of vegetation index (NDVI) to extract E. prolifera from moderate-resolution imaging spectroradiometer (MODIS) images in the Yellow Sea in 2016–2018. Hu et al. [15] proposed lower and upper bounds of floating algae index (FAI) method to detect E. prolifera areas from remote sensing images. Wu et al. [16] developed a semisupervised clustering method based on image spectral attributes to detect E. prolifera areas in MODIS images. Min et al. [17] used NDVI to extract and track E. prolifera areas in the Yellow Sea based on geostationary ocean color imager (GOCI) and Landsat images. Zhang et al. [18] applied NDVI to extract E. prolifera areas from huanjing-1A/1B (HJ-1A/1B), gaofen-1 (GF-1), and MODIS images.

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Fig. 1. Enteromorpha prolifera disasters.

Xu et al. [19] extracted and tracked E. prolifera areas in the Yellow Sea from MODIS images using the FAI and trajectory model. Hu et al. [20] extracted and inversed the biomass of E. prolifera areas from MODIS images using the FAI. As can be seen, the above methods mainly use NDVI, FAI and TAR to detect E. prolifera. The traditional E. prolifera detection method can achieve good results for remote sensing images detection under specific circumstances. However, due to the complex and variable shooting conditions of remote sensing satellites, this kind of method is difficult to adapt to the detection of E. prolifera in remote sensing images with large amounts of data under different circumstances. Its robustness and universality are poor.

The E. prolifera detection method based on CNN does not require manually designed features and can perform effective image feature learning and extraction based on its own unique network structure using massive data and labels [21], [22], [23], [24]. In addition, with sufficient training data, the method has good generalization ability and can maintain good robustness and universality under complex and variable conditions [25], [26], [27], [28], [29], [30], [31]. Some E. prolifera detection researches for remote sensing images based on CNN have been carried out. For example, Li et al. [32] developed an enteromorpha prolifera network (EPNet) model based on the U-Net framework for E. prolifera detection in MODIS images. Wang et al. [33] proposed an E. prolifera detection method for unmanned aerial vehicle (UAV) images by combining superpixels and CNN. First, the image was segmented to obtain multiscale patches. Then, the multiscale patches were classified to realize E. prolifera detection using CNN. Wan et al. [8] proposed 1-D-CNN and bidirectional long short-term memory (Bi-LSTM) networks considering sample balance to detect E. prolifera from GOCI images. Cui et al. [34] developed a super-resolution-based semantic segmentation network for E. prolifera extraction from MODIS images. Wang et al. [35] proposed an instance segmentation framework named AlgaeMask for E. prolifera detection from the surveillance videos. Compared with the traditional E. prolifera detection methods, E. prolifera detection methods based on CNN can achieve better results. However, due to the blurred boundary and diffuse distribution of E. prolifera areas in HSRIs, the existing E. prolifera detection methods based on CNN are usually difficult to accurately determine the boundary of E. prolifera areas.

For accurately detecting the boundary of E. prolifera areas in HSRIs, this article proposes an E. prolifera detection method for HSRIs based on boundary-assisted dual-path CNNs (BADP-CNNs). In this article, a large-scale HSRIs' E. prolifera detection dataset, FIO-EP, is created and published. Then, the BADP-CNN framework is designed to detect E. prolifera in HSRIs based on the shape distribution characteristics of E. prolifera. In the BADP-CNN framework, accurate detection of E. prolifera areas in HSRIs is achieved by fusing initial detection and boundary detection results of E. prolifera.

The following are the significant contributions of this article.

1) A large-scale HSRIs' E. prolifera detection dataset, FIO-EP, is created and published for the first time to facilitate the field of HSRIs E. prolifera detection.
2) The BADP-CNN framework is designed to accurately detect E. prolifera areas in HSRIs based on the shape distribution characteristics of E. prolifera.
3) E. prolifera areas with various distribution shapes under different conditions are accurately detected in HSRIs.

The rest of this article is structured as follows. Section II represents how to create an E. prolifera detection dataset and introduces how to construct, train, and test the BADP-CNN framework. Section III conducts, analyzes, and discusses the experimental findings. Section IV summarizes the experimental findings and draws conclusions.

II. Methodology

In this section, a detailed introduction to the FIO-EP dataset is first provided. Then, the expression and characteristics of the key stages of the BADP-CNN framework are represented.

A. FIO-EP Dataset Building

Enteromorpha prolifera usually goes through a development process of "emergence → development → outbreak → persistence → recession → extinction" for several months. At different stages of development, E. prolifera usually presents different distribution patterns [36]. Furthermore, E. prolifera has the characteristics of blurred boundary and diffuse distribution in the remote sensing images. At present, most of the research studies mainly detect E. prolifera from low- and medium-resolution remote sensing images (such as MODIS, GOCI, and HJ) [13], [14], [16]. Due to the low resolution of low- and medium-resolution remote sensing images, it is difficult to be effectively used for the detection of E. prolifera in the early stage and diffuse distribution of E. prolifera. HSRIs have a higher spatial resolution than low- and medium-resolution remote sensing images. Then, HSRIs

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TABLE I
PARAMETERS OF THE SATELLITE SENSORS

Parameter information	GF-1 B/C/D	GF-2	ZY3-02
Spectral band (μm)	Band 1: 0.45-0.52 Band 2: 0.52-0.59 Band 3: 0.63-0.69 Band 4: 0.77-0.89	Band 1: 0.45-0.52 Band 2: 0.52-0.59 Band 3: 0.63-0.69 Band 4: 0.77-0.89	Band 1: 0.45-0.52 Band 2: 0.52-0.59 Band 3: 0.63-0.69 Band 4: 0.77-0.89
Ground sample distance	8 m	4 m	5.8 m
Swath width	60 km	45 km	52 km
Radiometric resolution	10 bits	10 bits	10 bits
Revisit cycle	4 days	5 days	5 days

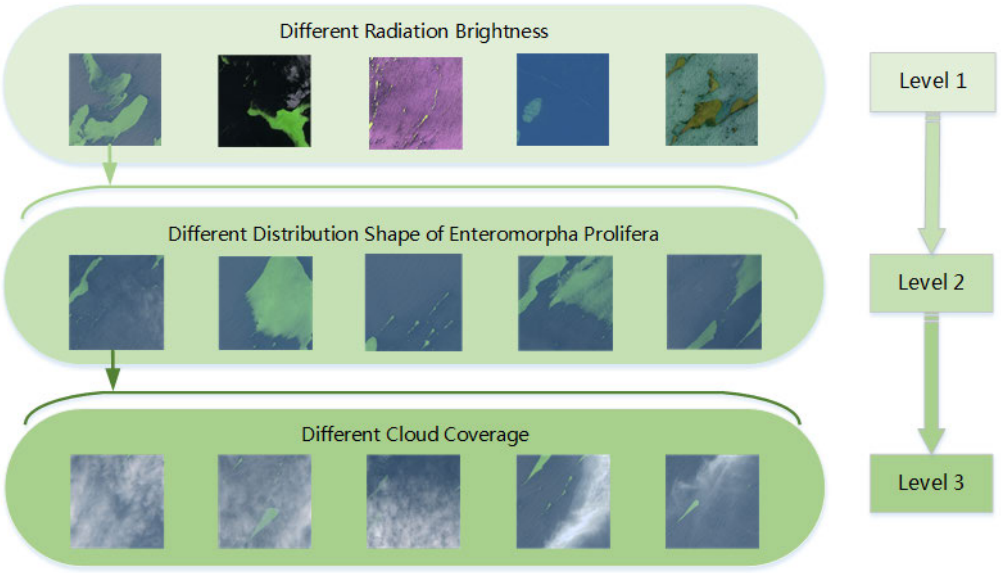


Fig. 2. Various types of E. prolifera HSRIs.

can be effectively used for the detection of E. prolifera in different distribution shapes at different development stages. Large-scale semantic segmentation datasets are the foundation and essential for the semantic segmentation frameworks based on CNN to achieve high performance [37]. However, there is no public HSRIs' E. prolifera semantic segmentation dataset. With respect to this problem, a large-scale HSRIs' E. prolifera detection dataset, FIO-EP, is created and published to facilitate the field of HSRIs' E. prolifera detection in this article.

To ensure the diversity and richness of E. prolifera image samples, multiple satellite sensor images (such as GF1 B/C/D, GF-2, and ZY3-02) are used to build the dataset. The parameters of the satellite sensors that are used in this article are shown in Table I. In this article, bands 1–3 of the HSRIs are selected to form a true color image for the detection of E. prolifera area. In the true color image, E. prolifera area appears to be green, which is obviously different from other features and is conducive to the extraction of E. prolifera areas. The experimental statistical analysis shows that radiation brightness, distribution shape of E. prolifera, and cloud coverage are the main factors that affect the accuracy of HSRIs' E. prolifera detection using semantic segmentation framework based on CNN. Therefore, the classification and collection of multisource HSRIs are guided from three aspects: radiation brightness, distribution shape of E. prolifera, and cloud coverage, as shown in Fig. 2. In Fig. 2, different species of E. prolifera HSRIs are first collected based on different radiation brightness. Second, E. prolifera HSRIs with the different distribution shape are collected on the basis of different radiation brightness. Third, E. prolifera HSRIs with

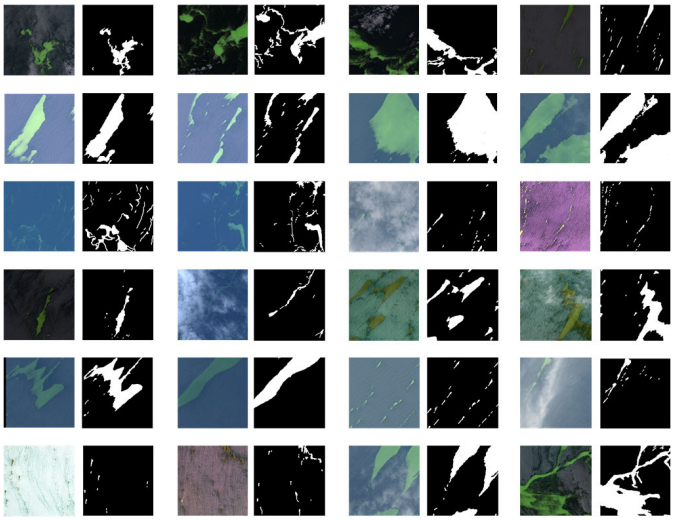


Fig. 3. Samples of the images and annotations in the FIO-EP dataset.

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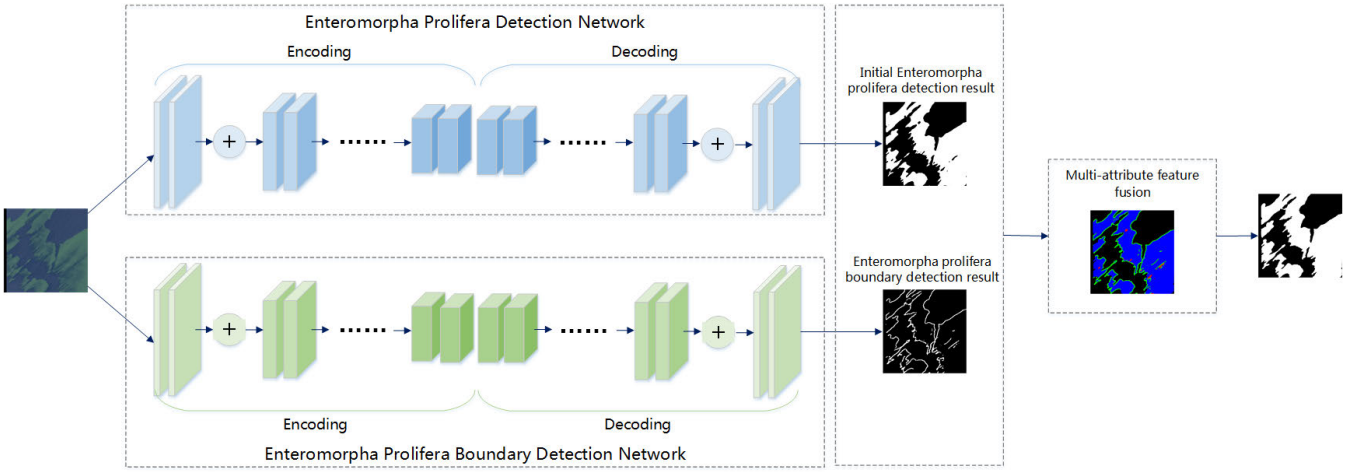


Fig. 4. Schematic of the BADP-CNN framework.

different cloud coverage are collected on the basis of different radiation brightness and distribution shape of E. prolifera. The collection and labeling of various types of E. prolifera HSRIs under different environmental conditions lay a good dataset foundation for the semantic segmentation framework based on CNN.

According to the classification criteria of different radiation brightness, cloud cover, and distribution shape of E. prolifera, E. prolifera HSRIs from different remote sensing satellites are collected and obtained. The obtained E. prolifera HSRIs are divided into 1334 image blocks, which are 512×512 pixels in size. Using the professional image semantic segmentation annotation tool, E. prolifera areas are accurately annotated in the 1334 image blocks by translation, rotating. and zooming operations to create the E. prolifera detection dataset, FIO-EP. Some samples of images and annotations in the FIO-EP dataset are shown in Fig. 3. In the annotations, the background and E. prolifera areas are identified as (0, 0, 0) and (255, 255, 255), respectively. To ensure a uniform distribution of the training data and test data, we randomly select 3/5, 1/5, and 1/5 images from the FIO-EP dataset as the training set, the verification set, and the test set, respectively.

B. Enteromorpha Prolifera Detection for HSRIs Based on CNN

1) BADP-CNN Framework Design: In HSRIs, E. prolifera areas have the characteristics of blurred boundary and diffuse distribution. However, the existing E. prolifera detection methods based on CNN are usually difficult to accurately determine the boundary of E. prolifera areas [38]. With respect to the problem, this article proposes an E. prolifera detection method for HSRIs based on BADP-CNNs. The schematic of the BADP-CNN framework is shown in Fig. 4. In Fig. 4, the initial E. prolifera detection result and E. prolifera boundary detection result are obtained using the dual-path CNN. By fusing the initial detection results of E. prolifera with high confidence and the boundary detection results of E. prolifera with good boundary dependence, accurate E. prolifera detection results can be obtained for HSRIs.

In the design of the BADP-CNN framework, the residual module is integrated into the network framework to prevent "gradient dispersion" and "gradient explosion" during the training process [39]. To ensure the scale symmetry between the network image input and the classification output, the advantage of scale symmetry between the encoding and decoding stages of the SegNet framework is fused into the BADP-CNN framework [40]. For more accurate object region extraction, the low-level location information features are fused into the high-level semantic information features in the BADP-CNN decoding stage [41]. In this article, the BADP-CNN framework takes full advantages of the residual module optimization training parameters, scale symmetry between the encoding and decoding stages of SegNet framework, and more accurate object region extraction by fusing the low-level location information features into the high-level semantic information features. The detailed design of the BADP-CNN framework is shown in Fig. 5. In Fig. 5, the BADP-CNN framework is a dual-path CNN containing the E. prolifera detection network and E. prolifera boundary detection network. The E. prolifera detection network and E. prolifera boundary detection network are used to obtain the initial detection results of E. prolifera and E. prolifera boundary detection results, respectively. The E. prolifera detection network is the same as E. prolifera boundary detection network. In the E. prolifera detection network, the image features are extracted using five levels of features in the encoding stage. There are two groups of feature maps and a residual module in each level of features. In the decoding stage, the image features are extracted using six levels of features. Among six levels of features, the first five levels of features contain two groups of feature maps and a residual module. Moreover, the feature information extraction in the decoding stage incorporates the feature maps processed by the encoding stage residual module. The sixth level of features contains a group of feature maps and a logistic layer. The logistic layer normalizes the output of the sixth level feature map to 0–1. The logistic function is displayed in (1). The logistic layer output values are used to compute the training loss of the network and forecast the

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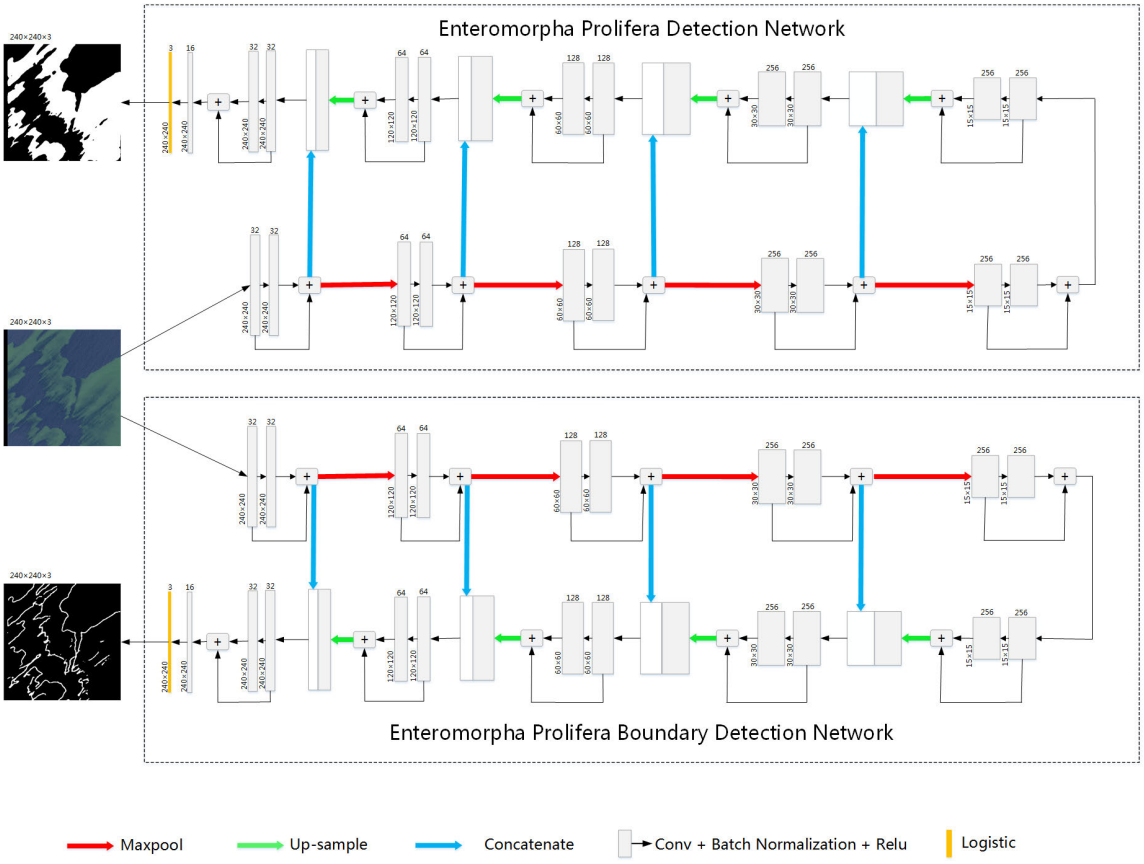


Fig. 5. BADP-CNN framework.

semantic segmentation results. In the E. prolifera detection network, the number of bands, height, and width of the input image are 3, 240, and 240, respectively. In the max-pool layers, the stride and size of the pool kernel are set to 2 and 2, respectively. The pad, stride, and size of the convolutional kernel in the convolutional layers are set to 1, 1, and 3, respectively. In this article, the BADP-CNN framework is trained using the stochastic gradient descent (SGD) [42]. During the training process, min-batch size, weight decay, and momentum are set to 10, 0.0001, and 0.9, respectively. The training rate for the first 50 000 min-batches is 0.0001 and for the last 50 000 min-batches is 0.00001. In the ground-truth annotations, the pixel of the background, E. prolifera area, and E. prolifera boundary are marked as (0, 0,0), (255, 255, 255), and (255, 255, 255), respectively. When training the BADP-CNN framework parameters, the pixel values in the ground-truth annotations need to be divided by 255 to normalize to 0–1. The training loss of the BADP-CNN is calculated based on the output values of the logistic layers in dual-path CNN and normalized ground-truth annotations. The training loss is calculated as in (1)–(5), shown at the bottom of the page, where x represents the output value of the feature map in the sixth level feature. $L(\{EP_{kijr}\}, \{EPB_{kijr}\})$ represents the training loss of the BADP-CNN framework.

$$s(x) = \frac{1}{1+e^{-x}} \tag{1}$$

$$L\left(\left\{\mathrm{EP}_{kijr}\right\}, \left\{\mathrm{EPB}_{kijr}\right\}\right) = \lambda L_{\mathrm{EPDN}}\left(\left\{\mathrm{EP}_{kijr}, \mathrm{EP}^{*}_{kijr}\right\}\right) + (1-\lambda) L_{\mathrm{EPBDN}}\left(\left\{\mathrm{EPB}_{kijr}, \mathrm{EPB}^{*}_{kijr}\right\}\right) \tag{2}$$

$$L_{\mathrm{EPDN}}\left(\left\{\mathrm{EP}_{kijr}, \mathrm{EP}^{*}_{kijr}\right\}\right) = \sqrt{\frac{1}{\omega \times m \times n \times 3} \sum_{k}^{\omega} \sum_{i}^{m} \sum_{j}^{n} \sum_{r}^{3} \left(\mathrm{EP}_{kijr} - \mathrm{EP}^{*}_{kijr}\right)^{2}} \tag{3}$$

$$L_{\mathrm{EPBDN}}\left(\left\{\mathrm{EPB}_{kijr}, \mathrm{EPB}^{*}_{kijr}\right\}\right) = \sqrt{\frac{1}{\omega \times m \times n \times 3} \sum_{k}^{\omega} \sum_{i}^{m} \sum_{j}^{n} \sum_{r}^{3} \left[\eta \times \left(\mathrm{EPB}_{kijr} - \mathrm{EPB}^{*}_{kijr}\right)^{2}\right]} \tag{4}$$

$$\eta = \begin{cases} 1 & \mathrm{EPB}^{*}_{kijr} = 1 \\ \alpha & \mathrm{EPB}^{*}_{kijr} \neq 1 \end{cases} \tag{5}$$

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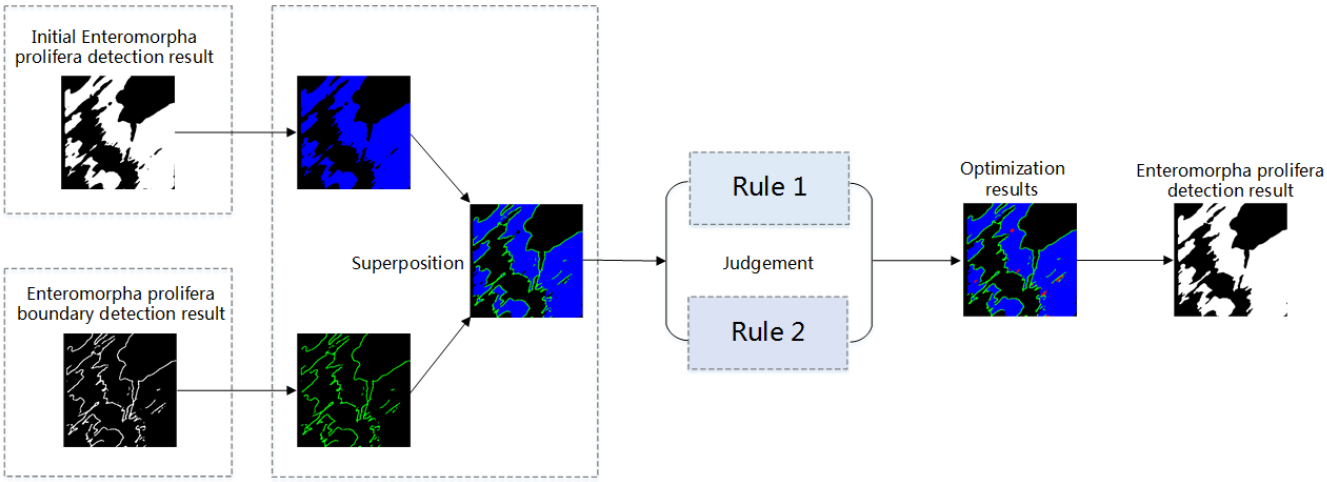


Fig. 6. Enteromorpha prolifera boundary-assisted E. prolifera detection.

$L_{EPDN}(\{EP_{kijr}, EP^*_{kijr}\})$ and $L_{EPBDN}(\{EPB_{kijr}, EPB^*_{kijr}\})$ represent the training loss of the E. prolifera detection network and E. prolifera boundary detection network, respectively. λ represents the balance weight of the training loss of the two networks ($\lambda = 0.4$). ω represents the number of images in the min-batch. n and m represent the width and height of the logistic layer in the sixth level feature, respectively. EP_{kijr} represents the output value of the logistic layer of the E. prolifera detection network at position (r,j,i,k). EP^*_{kijr} represents the truth value of the normalized ground-truth annotations at position (r, j, i, k), which is used to train the E. prolifera detection network. EPB_{kijr} represents the output value of the logistic layer of the E. prolifera boundary detection network at position (r, j, i, k). EPB^*_{kijr} represents the truth value of the normalized ground-truth annotations at position (r, j, i, k), which is used to train the E. prolifera boundary detection network. When training the E. prolifera boundary detection network, there is a large difference between the number of foreground pixels and the number of background pixels. Then η is a balance coefficient used to balance the ratio of foreground pixels to background pixels. When the pixel is the foreground, η is set to 1 and vice versa to α ($\alpha = 1/50$).

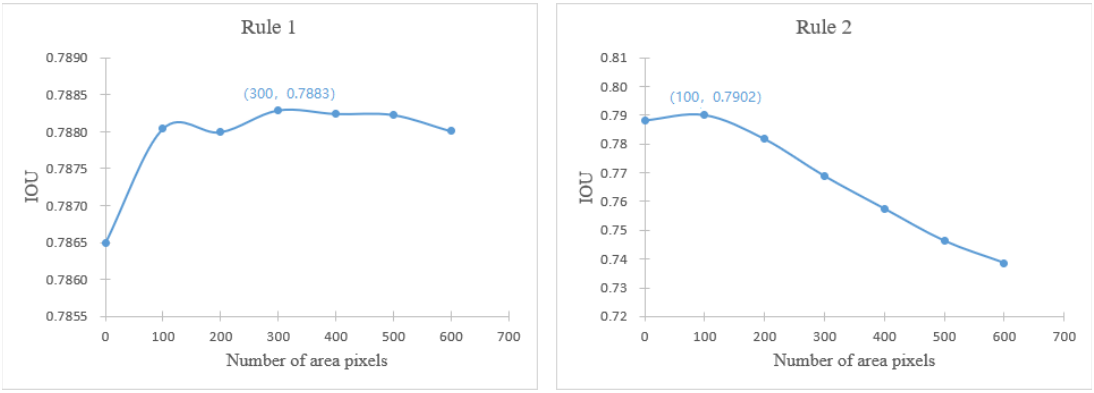
During the testing of the BADP-CNN framework, the output values of the two logistic layers in the dual-path CNN are multiplied by 255 to map to the range of 0–255. Based on the output values of the mapped logistic layers, two gray images are obtained using (6), which is the most popular equation for the three-band image to gray image [43]. If the pixel value in two gray images is not less than the set threshold, the predicted image pixel at the position matching to the gray image is E. prolifera or E. prolifera boundary, and vice versa is the background. In this article, the threshold is set to 200. Two predicted images containing the initial E. prolifera detection results and E. prolifera boundary detection results are obtained after the two gray image pixels are judged. In the two predicted images, the pixels are marked as (0, 0, 0), (255, 255, 255), and (255, 255, 255) in the background, E. prolifera areas, and boundaries, respectively,

$$V_{ij}^{\text{gray}} = B_{ij}^1 \times \rho_1 + B_{ij}^2 \times \rho_2 + B_{ij}^3 \times \rho_3 \tag{6}$$

where V_{ij}^{gray} represents the gray image pixel value at position (i, j). B_{ij}^1, B_{ij}^2, and B_{ij}^3 represent the output values of the mapped logistic layer three feature maps at position (i, j), respectively. ρ_1, ρ_2, and ρ_3 are the conversion coefficients, which are 0.299, 0.587, and 0.114, respectively.

2) Enteromorpha Prolifera Boundary-Assisted E. prolifera Detection: The predicted images of the initial E. prolifera detection result and E. prolifera boundary detection result can be obtained using the BADP-CNN framework. The initial E. prolifera detection result can accurately detect the obvious areas of E. prolifera in the image; however, it is difficult to detect the fuzzy and diffuse areas at the boundary of E. prolifera. The E. prolifera boundary detection result can accurately obtain the contour boundary of E. prolifera area in the image. Then, the E. prolifera boundary detection result is used to assist in optimizing the initial E. prolifera detection result to accurately extract E. prolifera area in the image. The process of optimizing the initial E. prolifera detection result with the assistance of the E. prolifera boundary detection result is shown in Fig. 6. In Fig. 6, the initial E. prolifera detection result and the E. prolifera boundary detection result are superposed first. In the superposition result, there are some E. prolifera area "gaps" and E. prolifera noise. In this article, Rules 1 and 2 are proposed to fill E. prolifera area "gaps" and eliminate E. prolifera noise, as shown below. After judging the superposition result using Rules 1 and 2, the initial E. prolifera detection result is optimized to obtain accurate E. prolifera detection result for HSRIs. Based on the experimental statistical analysis, the experimental results for different number threshold of area pixels in Rules 1 and 2 are shown in Fig. 7(a) and (b). In Fig. 7(a), the intersection-over-union (IOU) of the experimental results is optimal when the number threshold of area pixels in Rule 1 is set to 300. In Fig. 7(b), the IOU of the experimental results is optimal when the number threshold of area pixels in Rule 2 is set

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(a) (b)

Fig. 7. (a) and (b) IOU values corresponding to different numbers of area pixels in Rules 1 and 2, respectively.

to 100. Therefore, the number threshold of area pixels in Rules 1 and 2 is set to 300 and 100, respectively.

Rule 1: In the superposition result, regional growth is carried out based on the categories of E. prolifera and non-E. prolifera. If the number of pixels in a non-E. prolifera area is less than 300, the area is filled as E. prolifera area.

Rule 2: In the regional growth result, if the number of pixels in an E. prolifera area is less than 100, the E. prolifera region is noisy and becomes a non-E. prolifera area.

III. Experimental Results and Discussion

The performance of the BADP-CNN framework is fully verified by comparing and evaluation with some state-of-the-art semantic segmentation algorithms (e.g., EPNet [32], U-Net [41], and DeeplabV3+ [44]). Based on the Darknet framework, the BADP-CNN framework is programmed in C++. The experiments are carried out on a server with a Windows 10 operating system, an Inter CPU E5-2667 v4 at 3.20 GHz, an NVIDIA Quadro M4000 graphics card with 8 GB of GPU memory, and 16 GB of RAM.

A. Evaluation Criteria

In this article, qualitative evaluation and quantitative evaluation are used to evaluate the E. prolifera detection algorithms. Visual evaluation is the most widely used qualitative evaluation method. Through visual evaluation, we can directly observe the missing and false detection of the object detection results [45], [46]. Different semantic segmentation algorithms are evaluated quantitatively for their ability to detect E. prolifera using the F1-score, IOU and over-accuracy (OA) [47], [48]. The F1-score value lies between [0, 1]. The accuracy of the results of the E. prolifera detection algorithms increases with increasing F1-score value, and vice versa. The IOU value lies between [0, 1]. The consistency of the results of the E. prolifera detection algorithms with the ground-truth annotation increases with increasing IOU value, and vice versa. The OA value lies between [0, 1]. The classification accuracy of the E. prolifera detection algorithms increases with increasing OA value, and vice versa. The following equations are used to determine the F1-score, IOU, and OA, respectively,

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{7}$$

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{8}$$

$$\text{F1-score} = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \tag{9}$$

$$\text{IOU} = \frac{|\text{EP} \cap \text{EP}^*|}{|\text{EP} \cup \text{EP}^*|} \tag{10}$$

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \tag{11}$$

where *TP* and *FP* represent the number of correctly detected E. prolifera pixels and incorrectly detected E. prolifera pixels in the image, respectively. *TN* and *FN* represent the number of correctly detected background pixels and incorrectly detected background pixels in the image, respectively. *EP* represents E. prolifera pixel set in the E. prolifera detection results. EP^* represents E. prolifera pixel set in the ground-truth annotations. |*| denotes to count how many pixels are present in the set.

TABLE II
Quantitative Evaluation of the BADP-CNN Framework

	Precision	Recall	F1-score	IOU	OA
Baseline	**0.9688**	0.807	0.8805	0.7865	0.99
Baseline+R1	0.8396	**0.9281**	0.8816	0.7883	0.9907
Baseline+R1+R2	0.8503	0.9178	**0.8828**	**0.7902**	**0.991**

B. Quantitative Evaluation of the BADP-CNN Framework

Large-scale semantic segmentation dataset is the foundation and essential for the semantic segmentation framework based on CNN to achieve high performance. In this article, the FIO-EP dataset is used to test the performance of the BADP-CNN framework. In the FIO-EP dataset, there are 1334 image blocks, which are 512×512 pixels in size. From the FIO-EP dataset, 800, 267, and 267 images are randomly selected as the training set, the verification set, and the test set, respectively. Table II displays the quantitative evaluation findings for the BADP-CNN framework using the FIO-EP dataset.

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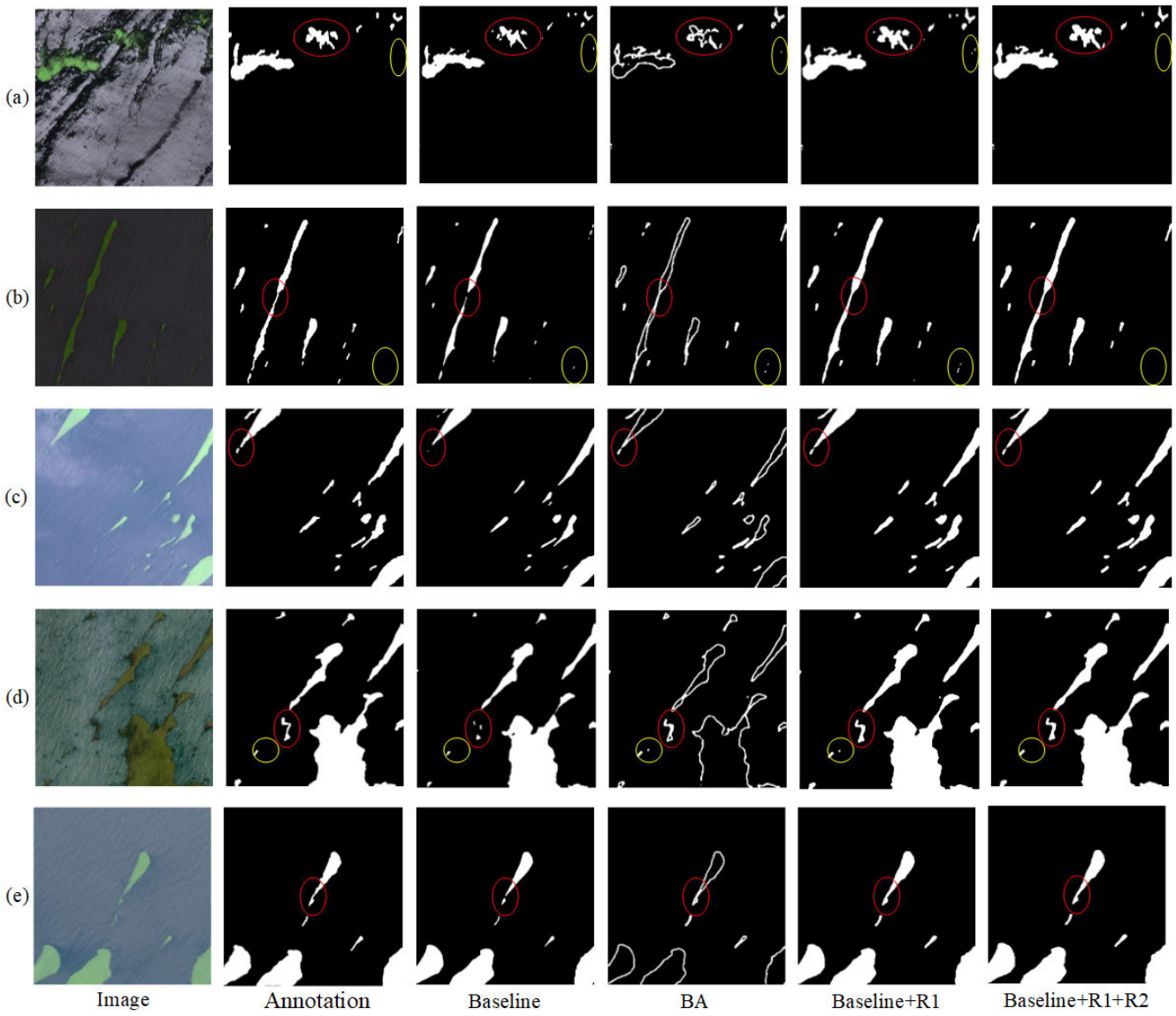


Fig. 8. Enteromorpha prolifera detection results using the BADP-CNN framework. (a)–(e) One to six columns represent the HSRIs, ground-truth annotations, initial Enteromorpha detection results, Enteromorpha boundary detection results, Enteromorpha optimization results using Rule 1, and Enteromorpha optimization results using the combination of Rules 1 and 2, respectively. The main comparison area of the detection results of the BADP-CNN framework is within red ovals and yellow ovals.

In Table II, the Baseline represents the initial E. prolifera detection results of the BADP-CNN framework for the FIO-EP dataset. The Baseline+R1 and Baseline+R1+R2 represent the optimization results of the initial Enteromorpha detection of the BADP-CNN framework for the FIO-EP dataset using Rule 1 and the combination of Rules 1 and 2, respectively.

In Table II, the F1-score of Baseline, Baseline+R1, and Baseline+R1+R2 is 0.8805, 0.8816, and 0.8828, respectively. The F1-score of Baseline+R1 and Baseline+R1+R2 is 0.11% and 0.23% higher than that of Baseline, respectively, indicating that more accurate E. prolifera detection results can be obtained using Rules 1 and 2 optimization. The IOU of Baseline, Baseline+R1, and Baseline+R1+R2 is 0.7865, 0.7883, and 0.7902, respectively. The IOU of Baseline+R1 and Baseline+R1+R2 is 0.18% and 0.37% higher than that of Baseline, respectively, showing that the E. prolifera detection results using Rules 1 and 2 optimization are more the same as the ground-truth annotations. The OA of Baseline, Baseline+R1, and Baseline+R1+R2 is 0.99, 0.9907, and 0.991, respectively. The OA of Baseline+R1 and Baseline+R1+R2 is 0.07% and 0.1% higher than that of Baseline, showing that Baseline+R1 and Baseline+R1+R2 have more accurate image classification results than Baseline. The above experimental result analysis shows that the boundary-assisted E. prolifera detection can achieve higher accuracy and is more suitable for E. prolifera detection of HSRIs.

C. Qualitative Evaluation of the BADP-CNN Framework

In this section, the E. prolifera detection results of the BADP-CNN framework are evaluated visually using FIO-EP dataset. Fig. 8 shows the detection results of different types of E. prolifera in multisource HSRIs using the BADP-CNN framework. In Fig. 8, Baseline, BA, Baseline+R1, and

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Baseline+R1+R2 represent the initial E. prolifera detection results, E. prolifera boundary detection results, optimization results of boundary-assisted initial Enteromorpha detection using Rule 1, and optimization results of boundary-assisted initial Enteromorpha detection using the combination of Rules 1 and 2, respectively. In Fig. 8(a)–(e), one to six columns represent the HSRIs, ground-truth annotations, initial Enteromorpha detection results, Enteromorpha boundary detection results, Enteromorpha optimization results using Rule 1, and Enteromorpha optimization results using the combination of Rules 1 and 2, respectively.

In Fig. 8(a), the diffuse distribution of E. prolifera within the red oval area cannot be detected accurately in the initial E. prolifera detection result. The diffusely distribution of E. prolifera area boundaries within the red oval area can be accurately detected in the optimization results of boundary-assisted initial Enteromorpha detection using Rule 1. However, E. prolifera anomalous noises within the yellow oval area are difficult to effectively suppress in the initial E. prolifera detection result and Enteromorpha optimization results using Rule 1. E. prolifera anomalous noises within the yellow oval area can be effectively suppressed in the Enteromorpha optimization results using the combination of Rules 1 and 2. By optimizing the initial E. prolifera detection result using Rules 1 and 2, the diffuse distribution of E. prolifera area within the red oval area can be accurately detected, and E. prolifera anomalous noises within the yellow oval area can be effectively suppressed. In Fig. 8(b), the fuzzy distributed E. prolifera within the red oval area is difficult to be detected accurately in the initial E. prolifera detection result. The fuzzy distributed E. prolifera area boundaries within the red oval area can be accurately detected in the E. prolifera boundary detection result. By optimizing the initial E. prolifera detection result using the boundary detection result based on Rule 1, the fuzzy distributed E. prolifera area within the red oval area can be accurately detected. However, the E. prolifera anomalous noises within the yellow oval area are difficult to effectively suppress in the initial E. prolifera detection result and Enteromorpha optimization results using Rule 1. The E. prolifera anomalous noises within the yellow oval area can be effectively suppressed in the Enteromorpha optimization results using the combination of Rules 1 and 2. In Fig. 8(c), the fuzzy distributed E. prolifera within the red oval area is difficult to be detected accurately in the initial E. prolifera detection result. In the E. prolifera boundary detection result, the fuzzy distributed E. prolifera area boundaries within the red oval area can be accurately detected. In the optimization result, the fuzzy distributed E. prolifera area within the red oval area can be accurately detected by optimizing the initial E. prolifera detection result based on the boundary detection result. In Fig. 8(d), the fuzzy distribution of E. prolifera within the red oval area is difficult to be detected accurately in the initial E. prolifera detection result. In the E. prolifera boundary detection result, the fuzzy distribution of E. prolifera area boundaries within the red oval area can be accurately detected. In the optimization result using Rule 1, the fuzzy distribution of E. prolifera area within the red oval area can be accurately detected by optimizing the initial E. prolifera detection result using the boundary detection result. However, the E. prolifera anomalous noises within the yellow oval area are difficult to effectively suppress in the Enteromorpha optimization results using Rule 1. The E. prolifera anomalous noises within the yellow oval area can be effectively suppressed in the Enteromorpha optimization results using the combination of Rules 1 and 2. In Fig. 8(e), E. prolifera within the red oval area is difficult to be detected accurately in the initial E. prolifera detection result. The E. prolifera area boundaries within the red oval area can be accurately detected in the E. prolifera boundary detection result. By optimizing the initial E. prolifera detection result based on the boundary detection result, the E. prolifera area within the red oval area can be accurately detected in the optimization result.

The analysis of the experimental findings indicates that the fuzzy and diffuse distribution of E. prolifera is difficult to be accurately detected in the initial E. prolifera detection results. Furthermore, there are some E. prolifera anomalous noises in the initial E. prolifera detection results. In Enteromorpha optimization results using Rule 1, the fuzzy and diffuse distribution of E. prolifera areas can be accurately detected. In Enteromorpha optimization results using the combination of Rules 1 and 2, it can effectively suppress E. prolifera anomalous noises and accurately detect different types of E. prolifera. Therefore, in the BADP-CNN framework, boundary-assisted E. prolifera detection can more accurately detect E. prolifera areas for HSRIs.

TABLE III
QUANTITATIVE COMPARISON OF DIFFERENT ENTEROMORPHA PROLIFERA DETECTION ALGORITHMS

	Precision	Recall	F1-score	IOU	OA
FCN	0.9175	0.5838	0.7136	0.5547	0.9807
U-Net	0.9404	0.703	0.8046	0.6731	0.9859
SegNet	**0.9808**	0.7413	0.8444	0.7307	0.9887
ResUNet-a	0.603	0.7449	0.6665	0.4998	0.9693
DeeplabV3+	0.8782	0.8675	0.8677	0.7663	0.9861
EPNet	0.9643	0.6878	0.8029	0.6707	0.9892
BADP-CNN	0.8503	**0.9178**	**0.8828**	**0.7902**	**0.991**

D. Quantitative Comparison of Different Enteromorpha Prolifera Detection Algorithms

The performance of the BADP-CNN framework is fully tested and verified by comparing and evaluation with some state-of-the-art E. prolifera detection algorithms using the FIO-EP dataset [32], [40], [41], [44], [49], [50]. Table III shows quantitative comparison results of different E. prolifera detection algorithms for the FIO-EP dataset. In Table III, the precisions of the seven E. prolifera detection algorithms are 0.9175, 0.9404, 0.9808, 0.603, 0.8782, 0.9643, and 0.8503, respectively. Except for ResUNet-a, the precision of the other six algorithms is larger than 0.85, indicating that the other six algorithms can identify E. prolifera in HSRIs with high precision. The recalls of the six E. prolifera detection algorithms are 0.5838, 0.703, 0.7413, 0.7449, 0.8675, 0.6878, and 0.9178, respectively. The recall of the BADP-CNN is much larger than that of other E. prolifera detection algorithms, indicating that this algorithm can more accurately distinguish E. prolifera

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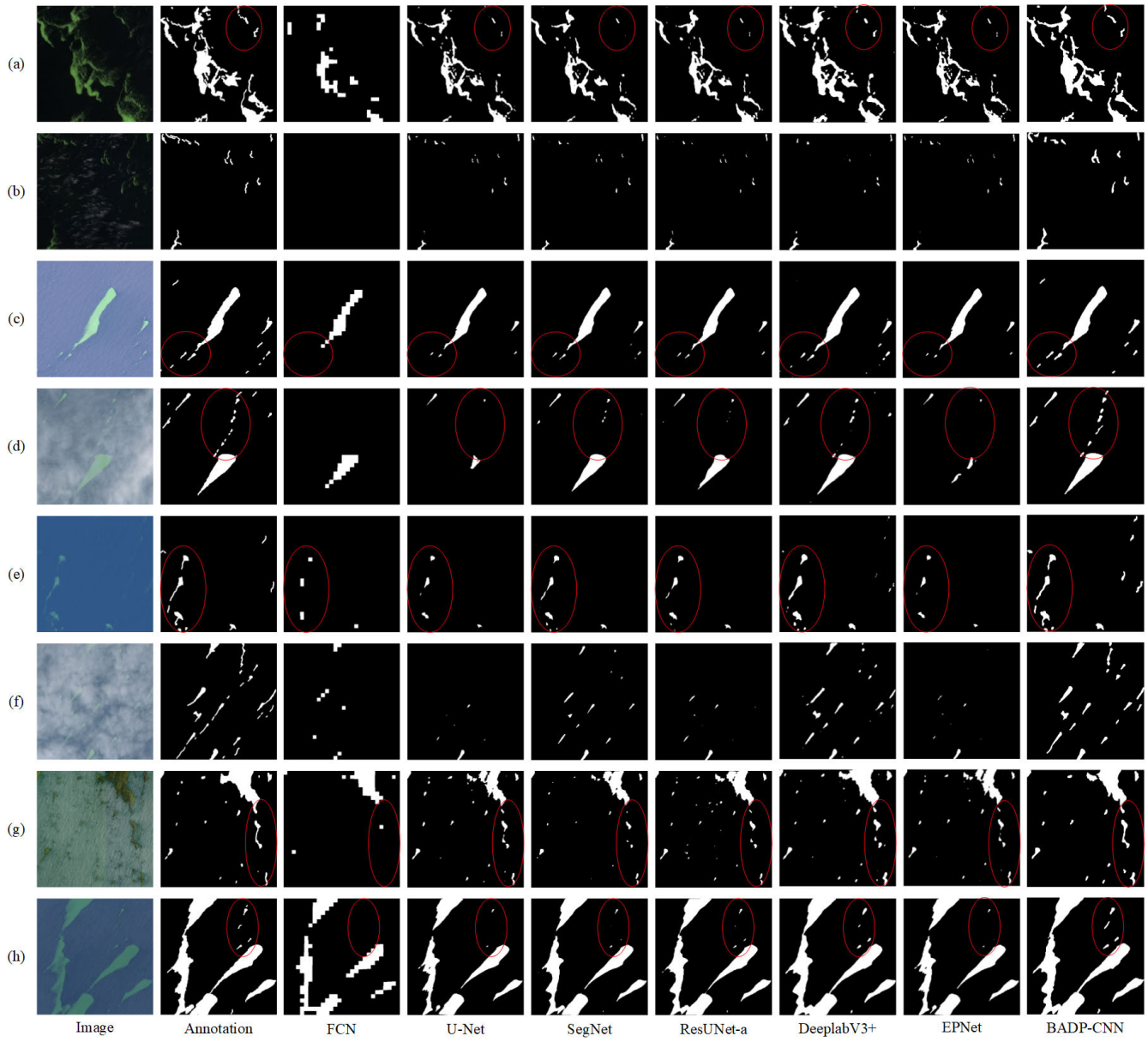


Fig. 9. Results of the seven E. prolifera detection algorithms for the FIO-EP dataset. (a)–(h) First two columns are the HSRIs and ground-truth annotations, respectively. The E. prolifera detection results of FCN, U-Net, SegNet, ResUNet-a, DeeplabV3+, EPNet, and BADP-CNN are displayed in the next columns, respectively. The main comparison area of the algorithm results is within red ovals.

from the background in HSRIs. The F1-scores of the seven E. prolifera detection algorithms are 0.7136, 0.8046, 0.8444, 0.6665, 0.8677, 0.8029, and 0.8828, respectively. The F1-score of BADP-CNN is the largest among the seven E. prolifera detection algorithms, indicating that BADP-CNN can detect E. prolifera in HSRIs more accurately. The IOUs of the seven E. prolifera detection algorithms are 0.5547, 0.6731, 0.7307, 0.4998, 0.7663, 0.6707, and 0.7902, respectively. The IOU of BADP-CNN is larger than that of the other six E. prolifera detection algorithms, showing that the E. prolifera detection results of BADP-CNN are more consistent with the ground-truth annotations. The OAs of the E. prolifera detection algorithms are 0.9807, 0.9859, 0.9887, 0.9693, 0.9892, 0.9861, and 0.991, respectively. The OA of BADP-CNN is larger than that of the other six E. prolifera detection algorithms, indicating that BADP-CNN has more accurate image classification results for HSRIs. The comparative analysis of experimental findings shows that the BADP-CNN outperforms other state-of-the-art E. prolifera detection algorithms and can achieve better E. prolifera detection results for HSRIs.

E. Evaluation of Model Parameters and Computational Complexity

The model parameter size and FLOPS of the seven E. prolifera detection algorithms are displayed in Table IV. The model parameter size and FLOPS represent the number of parameters and the number of times of multiply–adds in the E. prolifera detection algorithm, respectively. In Table IV, the model parameter size of the BADP-CNN framework is 46.4 MB, which is less than other six E. prolifera detection

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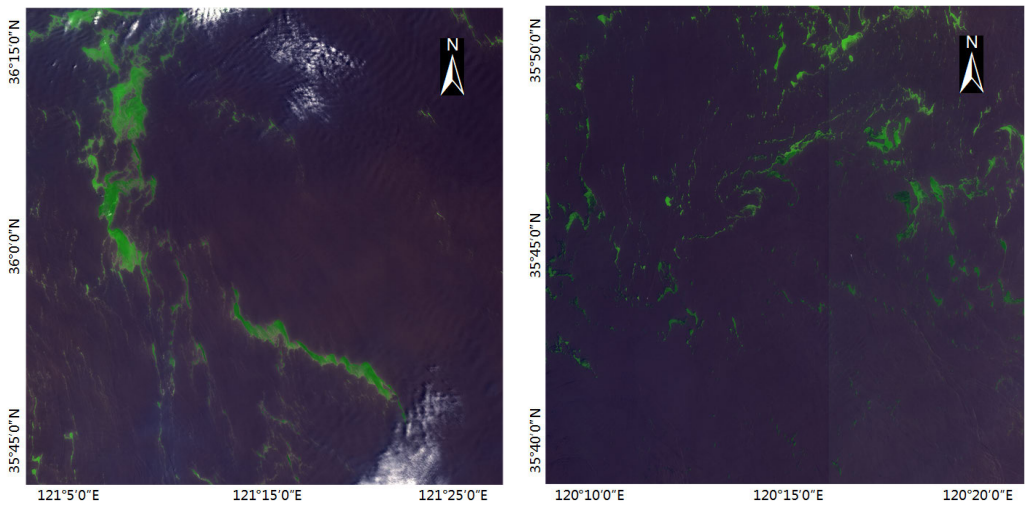


Fig. 10. (a) and (b) Large-scale GF-1B and ZY-3 images, respectively.

TABLE IV
EVALUATION OF MODEL PARAMETER AND COMPUTATIONAL COMPLEXITY OF DIFFERENT ENTEROMORPHA PROLIFERA ALGORITHMS

	Parameter (MB)	FLOPS
FCN	704	47.37 B
U-Net	131	114.89 B
SegNet	124	79.76 B
ResUNet-a	84.3	52.68 B
DeeplabV3+	59.2	88.54 B
EPNet	131	114.89 B
BADP-CNN	**46.4**	**46.93 B**

TABLE V
TIME CONSUMPTION COMPARISON OF DIFFERENT ENTEROMORPHA PROLIFERA DETECTION ALGORITHMS

	Time/Pre image (s)
FCN	0.1
U-Net	0.16
SegNet	0.12
ResUNet-a	0.12
DeeplabV3+	0.12
EPNet	0.15
BADP-CNN	0.1

algorithms, indicating that BADP-CNN has the least number of computational parameters. The FLOPS of the BADP-CNN framework is 46.93 B, which is the least among the seven E. prolifera algorithms, showing that BADP-CNN requires less multiply–adds operations. The comparative analysis of model parameters shows that BADP-CNN has the lowest model complexity and can obtain the optimal E. prolifera detection results for HSRIs using the least amount of computation.

F. Time Consumption Comparison

Table V shows the average time consumption of each image E. prolifera detection using the seven algorithms for the FIO-EP dataset. In Table V, the average time consumptions of the seven E. prolifera detection algorithms are 0.1, 0.16, 0.12, 0.12, 0.12, 0.15, and 0.1 s, respectively. The average time consumptions of the FCN and BADP-CNN are all 0.1 s, which is the least among the seven E. prolifera detection algorithms. The time consumption comparison findings indicate that the efficiency of BADP-CNN is better than SegNet, ResUNet-a, U-Net DeeplabV3+, and EPNet. The E. prolifera areas in HSRIs can be detected efficiently using the BADP-CNN framework.

G. Qualitative Comparison and Evaluation of the Enteromorpha Prolifera Detection Algorithms

To verify the performance of the BADP-CNN framework more comprehensively, the detection results of the BADP-CNN and other E. prolifera detection algorithms are compared and evaluated visually. Some samples of the results of the six E. prolifera detection algorithms for the FIO-EP dataset are shown in Fig. 9. In Fig. 9, the first two columns are the HSRIs and ground-truth annotations, respectively. The E. prolifera detection results of FCN, U-Net, SegNet, ResUNet-a, DeeplabV3+, EPNet, and BADP-CNN are displayed in the next columns, respectively.

In Fig. 9(a), the E. prolifera detection results of the FCN show a lot of false and missed detections compared with the ground-truth annotation. U-Net, ResUNet-a, SegNet, DeeplabV3+, and EPNet show missed detection of E. prolifera in red oval. The BADP-CNN can better detect E. prolifera in red oval, and its detection result is basically the same as the ground-truth annotation. In Fig. 9(b), FCN, U-Net, ResUNet-a, SegNet, DeeplabV3+, and EPNet show missed detection for E. prolifera with a small area of fuzzy and diffuse distribution. However, BADP-CNN can accurately detect E. prolifera with a small area of fuzzy and diffuse distribution. In Fig. 9(c), FCN, U-Net, ResUNet-a, SegNet, DeeplabV3+, and EPNet show missed detection for E. prolifera in red oval. BADP-CNN can better detect E. prolifera in red oval. In Fig. 9(d), it is difficult for the FCN to detect E. prolifera for the image with large amounts

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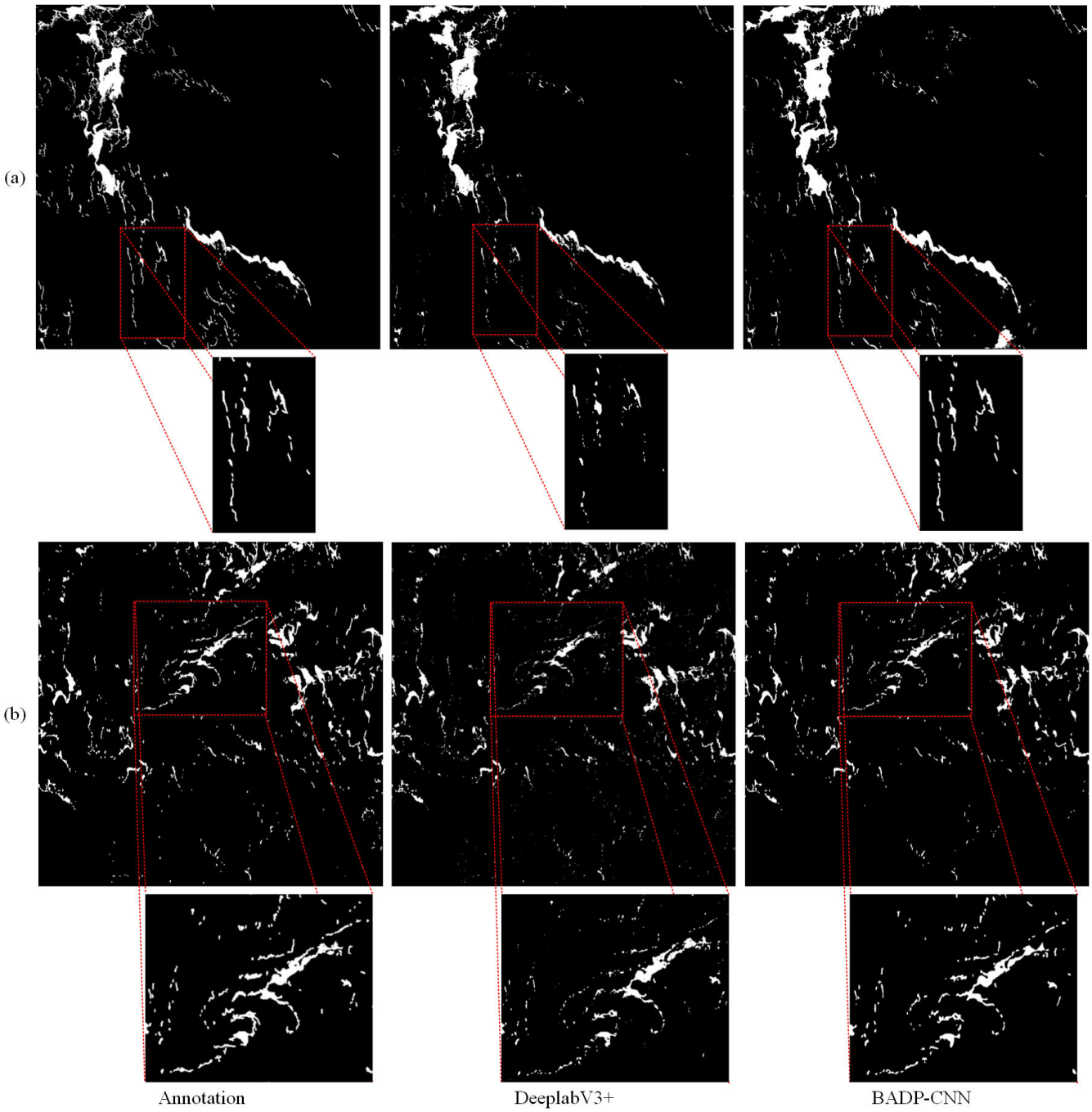


Fig. 11. (a) and (b) Three columns are the ground-truth annotations, E. prolifera detection results of DeeplabV3+, and E. prolifera detection results of BADP-CNN, respectively.

of cloud coverage. It is difficult for U-Net, ResUNet-a, SegNet DeeplabV3+, and EPNet to detect E. prolifera with the small area in red oval. BADP-CNN can better detect E. prolifera with the small area in red oval, and its detection result is basically the same as the ground-truth annotation. In Fig. 9(e), FCN, U-Net, ResUNet-a, SegNet, DeeplabV3+, and EPNet show missed detection for E. prolifera with a small area diffusely distributed in red oval. BADP-CNN can better detect E. prolifera with a small area diffusely distributed in red oval, and its detection result is basically the same as the ground-truth annotation. In Fig. 9(f), it is difficult for theFCN, U-Net, ResUNet-a, SegNet, DeeplabV3+, and EPNet to detect E. prolifera areas for the image with large amounts of cloud coverage. However, BADP-CNN can better detect E. prolifera areas from the image with large amounts of cloud coverage. In Fig. 9(g), it is difficult for the FCN to detect E. prolifera areas with a small area of diffusely distributed. U-Net, ResUNet-a, SegNet, DeeplabV3+, and EPNet show missed detection for E. prolifera in red oval. However, BADP-CNN can better detect E. prolifera in red oval, and its detection result is basically the same as the ground-truth annotation. In Fig. 9(h), it is difficult for the

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FCN, U-Net, ResUNet-a, SegNet, DeeplabV3+, and EPNet to detect E. prolifera with a small area in red oval. However, BADP-CNN can better detect E. prolifera with a small area in red oval.

The visual comparison and evaluation findings indicate that BADP-CNN outperforms the other six algorithms in E. prolifera detection of different types of HSRIs in the FIO-EP dataset. In particular, the E. prolifera detection results of BADP-CNN are significantly better than the other six E. prolifera detection algorithms for images with large amounts of cloud coverage. BADP-CNN can accurately detect small, fuzzy, and diffusely distributed E. prolifera areas in different types of HSRIs.

H. Enteromorpha Prolifera Detection for the Large-Scale HSRIs

To verify the robustness and universality of the BADP-CNN framework, two large-scale HSRIs are used to compare BADP-CNN with other E. prolifera detection algorithms. The two large-scale HSRIs are taken by satellite GF-1B and satellite ZY-3 on the Yellow Sea of China, as shown in Fig. 10(a) and (b). The size of two large-scale HSRIs is 5000×5000 pixels. Since the size of the large-scale HSRIs is large for the E. prolifera detection algorithms based on CNN, the large-scale HSRIs are divided into some contiguous 512×512 pixels sized image blocks for E. prolifera detection. After each image block is detected, the detection results of each image block are mapped into the original image to obtain the E. prolifera detection results for the large-scale HSRIs.

Table VI shows quantitative comparison results of seven E. prolifera detection algorithms for two large-scale HSRIs. In Table VI, the F1-score of the BADP-CNN is 0.8008, which is the largest among the seven E. prolifera detection algorithms, indicating that the BADP-CNN can detect E. prolifera in large-scale HSRIs more accurately. The IOU of BADP-CNN is 0.6678, which is larger than that of the other six E. prolifera detection algorithms, showing that the E. prolifera detection results of BADP-CNN are more consistent with the ground-truth annotations. The OA of BADP-CNN is 0.9813, which is larger than that of the other six E. prolifera detection algorithms, indicating that BADP-CNN has more accurate image classification results for HSRIs. The experimental findings indicate that BADP-CNN outperforms other E. prolifera detection algorithms and can achieve good E. prolifera detection results for large-scale HSRIs. Therefore, the BADP-CNN framework has good robustness and universality for E. prolifera detection of different types of images.

I. Visual Evaluation of Enteromorpha Prolifera Detection of the Large-Scale HSRIs

In this section, the E. prolifera detection results of large-scale HSRIs are visually evaluated. In Table VI, except for BADP-CNN, DeeplabV3+ has the best E. prolifera detection results for large-scale HSRIs among the other six algorithms. Therefore, the E. prolifera detection results of BADP-CNN are evaluated visually in comparison to those of DeeplabV3+. The E. prolifera detection results of DeeplabV3+ and BADP-CNN for the large-scale HSRIs are shown in Fig. 11. In Fig. 11(a), from the overall comparison, the E. prolifera detection results of BADP-CNN are more consistent with the ground-truth annotation, and are superior to that of DeeplabV3+. For E. prolifera within red rectangle, it can be seen that BADP-CNN outperforms DeeplabV3+ in detecting diffuse E. prolifera due to the integration of the boundary property features of E. prolifera. In Fig. 11(b), compared with the ground-truth annotation, the E. prolifera detection results of DeeplabV3+ have some missed detection. The E. prolifera detection results of BADP-CNN are basically the same as the ground-truth annotation. For E. prolifera within red rectangle, it can be seen that it is difficult for DeeplabV3+ to detect the boundary blurred and small-scale E. prolifera areas, while BADP-CNN can accurately detect these E. prolifera areas.

TABLE VI
QUANTITATIVE COMPARISON OF LARGE-SCALE HSRIS' ENTEROMORPHA PROLIFERA DETECTION RESULTS OF DIFFERENT E. PROLIFERA DETECTION ALGORITHMS

	Precision	Recall	F1-score	IOU	OA
FCN	0.9029	0.5119	0.6534	0.4852	0.9734
U-Net	**0.9704**	0.548	0.7004	0.539	0.9771
SegNet	0.9531	0.6075	0.742	0.5898	0.9793
ResUNet-a	0.9559	0.588	0.7281	0.5725	0.9785
DeeplabV3+	0.895	0.689	0.7786	0.6374	0.9808
EPNet	0.9597	0.5452	0.6954	0.533	0.9766
BADP-CNN	0.8369	**0.7678**	**0.8008**	**0.6678**	**0.9813**

The visual comparison evaluation findings indicate BADP-CNN outperforms DeeplabV3+ for E. prolifera detection of large-scale HSRIs. BADP-CNN can accurately detect diffuse, boundary blurred, and small-scale E. prolifera areas and obtain good E. prolifera detection results for large-scale HSRIs.

IV. CONCLUSION

In this article, an E. prolifera detection framework based on BADP-CNNs is designed to detect E. prolifera of HSRIs. With respect to the difficulty in accurate detection of E. prolifera area boundary in HSRIs, BADP-CNN detects E. prolifera by fusing the initial detection results of E. prolifera with high confidence and the boundary detection results of E. prolifera with good boundary dependence. Moreover, BADP-CNN takes full advantages of the residual module optimization training parameters, scale symmetry between the encoding and decoding stages of the SegNet framework, and more accurate object region extraction by fusing the low-level location information features into the high-level semantic information features. In this article, a large-scale HSRIs' E. prolifera detection dataset, FIO-EP, is created and published for the first time to facilitate the field of HSRIs' E. prolifera detection. To verify the performance of BADP-CNN, the FIO-EP dataset and two large-scale HSRIs are used to quantitatively and qualitatively compare BADP-CNN with some state-of-the-art E. prolifera detection algorithms. The experimental findings indicate that BADP-CNN outperforms other state-of-the-art E. prolifera detection algorithms. In particular, the E. prolifera detection results of BADP-CNN are significantly better than other

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state-of-the-art E. prolifera detection algorithms for images with large amounts of cloud coverage. BADP-CNN can accurately detect various distribution shapes of E. prolifera in different types of HSRIs.

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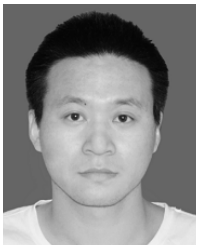
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